

COMPUTER NETWORKS — QUIZ PREPARATION

Chapter 1: Introduction · Layering · Protocols & Standards · Internet History · Line Coding · Error Detection

Name: _____ Date: _____ Score: _____ / 100

PART I — IMPORTANT TOPICS (REVIEW SHEET)

1. Layered models

TCP/IP layer	PDU	Addressing	Example protocols	Job
5 Application	Message	— (names)	HTTP, SMTP, DNS, FTP	Services to the user
4 Transport	Segment / datagram	Port	TCP, UDP	Process-to-process delivery
3 Network	Datagram / packet	IP (logical)	IP, ICMP	Host-to-host routing
2 Data link	Frame	MAC (physical)	Ethernet, Wi-Fi	Hop-to-hop (node to node)
1 Physical	Bits	—	Cables, signals	Bits onto the medium

- **OSI (7):** Application, Presentation, Session, Transport, Network, Data link, Physical. TCP/IP merges the top three into Application. (Mnemonic: "All People Seem To Need Data Processing".)
- **Encapsulation:** each layer adds its header (link layer also a trailer) to the PDU from above. **Decapsulation** at the receiver, bottom-up.
- **Peer-to-peer logical communication:** layer n talks to layer n on the other side; physically data goes down, across, up.
- **Devices:** hub/repeater = L1; switch = L2 (links); router = L3 (up to network layer); hosts = all 5 layers.
- **Overhead ratio** = header bytes / total bytes sent (or header/data — state which).

2. Protocols & standards

- **Protocol** = set of rules; key elements: **Syntax** (format), **Semantics** (meaning), **Timing** (when & how fast).
- **De facto** (by practice) vs **de jure** (by official body).
- **Bodies:** ISO, ITU-T, ANSI, IEEE (802.x LANs), EIA; Internet: **IETF** (writes RFCs), IAB, ISOC, IRTF.
- **RFC maturity:** Proposed → Draft → Internet Standard (plus Historic, Experimental, Informational).

3. Networks — basics

- **Data communication criteria:** Delivery, Accuracy, Timeliness, Jitter. **Components:** message, sender, receiver, medium, protocol.
- **Data flow:** simplex (one way), half-duplex (both, one at a time), full-duplex (both simultaneously).
- **Network criteria:** performance (throughput, delay), reliability, security.
- **Connection:** point-to-point vs multipoint. **Topologies:** mesh ($n(n-1)/2$ links, $n-1$ ports each), star, bus, ring.
- **LAN / WAN / MAN;** internetwork = network of networks; **Internet** = hierarchy of ISPs (tier 1 backbone, regional, local).
- **Switching:** circuit-switched (reserved resources, telephone) vs packet-switched (store & forward, Internet).

4. Internet history (key dates)

Year	Event
1961–64	Kleinrock / Baran: packet switching theory
1969	ARPANET — first 4 nodes (UCLA, SRI, UCSB, Utah)
1972	Email; NCP protocol
1973–74	Cerf & Kahn: TCP (internetworking)
1983	ARPANET switches to TCP/IP; DNS
1986	NSFNET
1990–91	Berners-Lee: World Wide Web (HTTP, HTML)
1995+	Commercialization, ISPs

5. Line coding

- **Unipolar NRZ:** 1 = +V, 0 = 0 V, whole bit time. Has DC component; no self-clocking.
- **Polar NRZ-L:** level = bit. **NRZ-I:** 1 = invert, 0 = no change.
- **Manchester:** transition in the middle of every bit (self-clocking, no DC, 2× bandwidth). IEEE 802.3: 0 = high → low, 1 = low → high (G.E. Thomas is the reverse — state convention).
- **Differential Manchester:** always mid-bit transition; 0 = transition at start, 1 = no transition at start.

6. Error detection

- **Single parity:** detects any odd number of bit errors; misses all even numbers. Even parity: total 1s even; odd parity: total 1s odd.
- **2-D parity:** row + column parity; detects all 1-, 2-, 3-bit errors, can correct single-bit errors; misses some 4-bit (rectangle) errors.
- **Checksum (1's complement):** add words, wrap carries back in, complement → checksum. Receiver: sum everything incl. checksum, complement → 0 means OK.
- **Number of k-bit errors in n-bit word:** C(n,k).

PART II — PRACTICE QUIZ

1. Represent the bit string **10110010** as (a) unipolar NRZ and (b) Manchester (IEEE 802.3). [10]

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2. Represent **01101001** as unipolar NRZ and Manchester. [10]

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3. Show the 2-D parity of **00010101**, **10110011**, **01000001** using **odd** parity. [10]

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4. Show the 2-D parity of **11001010**, **01110001**, **10100110** using **even** parity. [10]

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5. Compute the 8-bit one's-complement checksum for the codes of Question 3. Verify it at the receiver. [10]

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6. Compute the 8-bit checksum for **11001100**, **10101010**, **11110000**. [10]

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7. For the string **1001P**, where P is the **odd** parity bit, which two-bit errors are not detected? List all resulting strings. [10]

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8. For **1101P** with **even** parity: how many 1-, 2-, and 3-bit errors are possible, and which are detected? [5]

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9. In the TCP/IP model, every layer that encapsulates adds a 20B header. Give the overhead ratio for sending (a) 1B and (b) 100B of application data. [5]

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10. Map the TCP/IP layers onto an autonomous walking robot (or a postal-letter analogy). [5]

11. Multiple choice. [5]

- (a) The PDU at the data link layer is a: i) segment ii) packet iii) frame iv) message
- (b) Which organization publishes RFCs? i) IEEE ii) IETF iii) ITU-T iv) ANSI
- (c) Port numbers are used at the: i) network ii) transport iii) link iv) physical layer
- (d) A router operates up to layer: i) 1 ii) 2 iii) 3 iv) 5
- (e) Which line code always has a mid-bit transition? i) NRZ-L ii) NRZ-I iii) Unipolar NRZ iv) Manchester

12. Short answer: (a) three elements of a protocol; (b) de facto vs de jure; (c) links in a fully connected mesh of 6 devices; (d) circuit vs packet switching. [10]

PART III — ANSWER KEY

1. 10110010

bit : 1 0 1 1 0 0 1 0
NRZ : +V 0 +V +V 0 0 +V 0
Manch: L H H L L H L H H L L H H L (first half / second half)

NRZ



2. 01101001

bit : 0 1 1 0 1 0 0 1
NRZ : 0 +V +V 0 +V 0 0 +V
Manch: H L L H L H H L L H H L L H

Note: in Manchester there is an extra transition at the bit boundary between two equal bits.

3. Odd parity:

	Data	# of 1s	P
R1	0 0 0 1 0 1 0 1	3	0
R2	1 0 1 1 0 0 1 1	5	0
R3	0 1 0 0 0 0 0 1	2	1
Col P	0 0 0 1 1 0 0 0		1*

*Corner: parity of the column-parity row (2 ones) $\rightarrow$ 1; parity of the row-parity column (1 one) $\rightarrow$ 0. With odd parity they disagree because the number of rows (3) and columns (8) differ in parity — **state the convention you used.**

4. Even parity:

	Data	# of 1s	P
R1	1 1 0 0 1 0 1 0	4	0
R2	0 1 1 1 0 0 0 1	4	0
R3	1 0 1 0 0 1 1 0	4	0
Col P	0 0 0 1 1 1 0 1		0

(Even parity: the corner always agrees both ways.)

5.

00010101 (21)
+ 10110011 (179)
+ 01000001 (65)
= 1 00001001 (265) $\rightarrow$ wrap carry: 00001001 + 1 = 00001010
Complement $\rightarrow$ CHECKSUM = 11110101 (245)
Receiver: 21+179+65+245 = 510 $\rightarrow$ 11111110 + 1 (wrap) = 11111111 $\rightarrow$ complement = 000

6.

11001100 (204) + 10101010 (170) = 1 01110110 -> wrap -> 01110111 (119)
 01110111 (119) + 11110000 (240) = 1 01100111 -> wrap -> 01101000 (104)
 Complement -> CHECKSUM = 10010111 (151)
 Check: 104 + 151 = 255 = 11111111 -> complement 0 OK

7. 1001 has two 1s $\rightarrow P = 1 \rightarrow$ codeword **10011**. A single parity bit misses every 2-bit error: $C(5,2) = 10$ undetected strings:

01011, 00111, 00001, 00010, 11111, 11001, 11010, 10101, 10110, 10000 (flipping positions 1-2, 1-3, 1-4, 1-5, 2-3, 2-4, 2-5, 3-4, 3-5, 4-5). All have an odd number of 1s, so all pass.

8. 1101 has three 1s $\rightarrow P = 1 \rightarrow$ codeword 11011. 1-bit: $C(5,1)=5$, all detected. 2-bit: $C(5,2)=10$, none detected. 3-bit: $C(5,3)=10$, all detected.

9. Headers added by transport, network, link = $3 \times 20 = 60\text{B}$ (physical adds none).

(a) 1B: overhead = $60 / (60+1) = \mathbf{60/61 \approx 0.984}$ (or 60:1 relative to data).

(b) 100B: overhead = $60 / 160 = \mathbf{0.375}$ (or 60:100 = 0.6 relative to data).

If the application layer also adds 20B: $80/81 \approx 0.988$ and $80/180 \approx 0.444$. **State your assumption.**

10. Example robot mapping:

Layer	Robot walking
Application	Goal / intent: "walk to the door" (path planning)
Transport	Gait control: breaks the goal into steps, sequencing, checks each step completed (feedback / reliability)
Network	Balance & routing of movement: chooses which leg/joint path, foot placement, coordination
Data link	Joint-level controllers: commands to each motor, error checking of sensor/actuator messages
Physical	Motors, wires, signals, sensors, actual leg movement on the ground

11. (a) iii frame (b) ii IETF (c) ii transport (d) iii 3 (e) iv Manchester

12. (a) Syntax, semantics, timing. (b) De facto = adopted through widespread use; de jure = legislated by an official recognized body. (c) $n(n-1)/2 = 6 \cdot 5/2 = \mathbf{15}$ links. (d) Circuit: dedicated path reserved for the whole call (phone); packet: data split into packets, store-and-forward, resources shared on demand (Internet).

Last-minute checklist: state Manchester convention · wrap carries in checksum then complement · parity misses even # of errors · $C(n,k)$ for counting errors · overhead as a ratio with your assumption stated · PDU names: message $\rightarrow$ segment $\rightarrow$ datagram $\rightarrow$ frame $\rightarrow$ bits.